Be2C_moderator

November 2, 2026

[3]:
```
#Eliyahu Venarchick
#10/31/2026
#OpenMc Be2C Moderator, NO clad, 19.75%wt percentage U235

#␣
 ↪==========================================================================================
# Initialization
#␣
 ↪==========================================================================================

import os
import openmc
#import matplotlib.pyplot as plt



library_path = "/home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-viii.1-hdf5/
 ↪cross_sections.xml"
os.environ['OPENMC_CROSS_SECTIONS'] = library_path

#␣
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# Materials
#␣
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fuel = openmc.Material(0,name = "UO2") #Fuel Enriched 19.75 wt% U235
fuel.add_nuclide("U234",0.0001,'ao')
fuel.add_nuclide("U235",0.1975,'ao')
fuel.add_nuclide("U238",0.8124,'ao')
fuel.set_density('g/cm3',10.313) #Change Density Later to match with paper,␣
 ↪link: https://www.tandfonline.com/doi/epdf/10.1080/00295639.2022.2035157?
 ↪needAccess=true
fuel.add_element('O',2.0,'ao')


moderator = openmc.Material(name = 'Be2C')
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moderator.set_density('g/cm3',1.90)
moderator.add_element('Be',2.0)
moderator.add_element('C',1.0)
moderator.add_s_alpha_beta("c_Be_in_Be2C")

materials = openmc.Materials([fuel,moderator])

materials.cross_sections = library_path
materials.export_to_xml()

#␣
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# Geometry
#␣
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radius = 2.2
pitch = 5
fuel_OR = openmc.ZCylinder(r = radius, name = "Fuel Outer Radius")
box = openmc.model.RectangularPrism(pitch,pitch,boundary_type ='reflective')

pincell_universe = openmc.Universe(name = 'Fuel Pin Universe')

fuel_cell = openmc.Cell(name = "Fuel Cell",fill = fuel, region = -fuel_OR)
pincell_universe.add_cell(fuel_cell)

mod_cell = openmc.Cell(name = 'Moderator Cell',fill = moderator, region =␣
 ↪+fuel_OR)
pincell_universe.add_cell(mod_cell)

root_cell = openmc.Cell(name = 'Root Cell',fill = pincell_universe)
root_cell.region = -box

root_universe = openmc.Universe(universe_id=1,name = "Root Universe")
root_universe.add_cell(root_cell)

geometry = openmc.Geometry(root_universe)
geometry.export_to_xml()

#␣
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# Settings
#␣
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settings = openmc.Settings()

settings.batches = 150
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settings.inactive = 10
settings.particles = 5000

bounds = [-pitch/2, -pitch/2, -pitch/2, pitch/2, pitch/2, pitch/2]
uniform_dist = openmc.stats.Box(bounds[:3],bounds[3:],only_fissionable=True)
settings.source = openmc.IndependentSource(space = uniform_dist)
settings.export_to_xml()

#␣
 ↪==========================================================================================
#root_universe.plot(width =(pitch,pitch),basis='xy',colors={fuel_cell:␣
 ↪'green',mod_cell:'blue'})
#plt.title(f'radius = {radius}, pitch = {pitch}')

openmc.run()
```
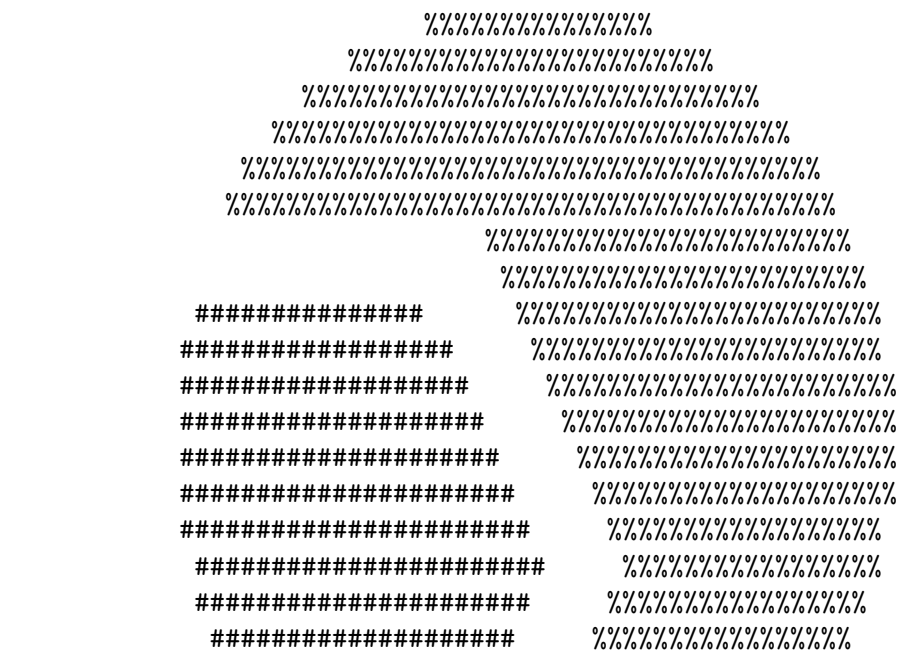

```
/home/n4sa_fan/anaconda3/envs/openmc-env/lib/python3.14/site-
packages/openmc/mixin.py:72: IDWarning: Another Material instance already exists
with id=0.
  warn(msg, IDWarning)
/home/n4sa_fan/anaconda3/envs/openmc-env/lib/python3.14/site-
packages/openmc/mixin.py:72: IDWarning: Another UniverseBase instance already
exists with id=1.
  warn(msg, IDWarning)
/home/n4sa_fan/anaconda3/envs/openmc-env/lib/python3.14/site-
packages/openmc/stats/multivariate.py:1115: FutureWarning: The
'only_fissionable' has been deprecated. Use the 'constraints' argument when
defining a source instead.
  warn("The 'only_fissionable' has been deprecated. Use the "
```



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                  ##################      %%%%%%%%%%%%%%%%%
                   ################      %%%%%%%%%%%%%%%%
                     ############      %%%%%%%%%%%%%%%
                        ########      %%%%%%%%%%%%%%
                                     %%%%%%%%%%

                 | The OpenMC Monte Carlo Code
       Copyright | 2011-2026 MIT, UChicago Argonne LLC, and contributors
         License | https://docs.openmc.org/en/latest/license.html
         Version | 0.16.0
     Commit Hash | 617d35a5063c57796b43428bc401e627d2011046
       Date/Time | 2026-11-02 08:20:16
  OpenMP Threads | 12

 Reading settings XML file…
 Reading cross sections XML file…
 Reading materials XML file…
 Reading geometry XML file…
 Reading U234 from
 /home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-viii.1-hdf5/neutron/U234.h5
 Reading U235 from
 /home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-viii.1-hdf5/neutron/U235.h5
 Reading U238 from
 /home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-viii.1-hdf5/neutron/U238.h5
 Reading O16 from
 /home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-viii.1-hdf5/neutron/O16.h5
 Reading O17 from
 /home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-viii.1-hdf5/neutron/O17.h5
 Reading O18 from
 /home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-viii.1-hdf5/neutron/O18.h5
 Reading Be9 from
 /home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-viii.1-hdf5/neutron/Be9.h5
 Reading C12 from
 /home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-viii.1-hdf5/neutron/C12.h5
 Reading C13 from
 /home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-viii.1-hdf5/neutron/C13.h5
 Reading c_Be_in_Be2C from
 /home/n4sa_fan/openmc/Cross_Section_Libraries/endfb-
viii.1-hdf5/thermal/c_Be_in_
 Be2C.h5
 Minimum neutron data temperature: 294 K
 Maximum neutron data temperature: 294 K
 Preparing distributed cell instances…
 Reading plot XML file…
 Writing summary.h5 file…
 Maximum neutron transport energy: 20000000 eV for O17
 Initializing source particles…
```

==========> K EIGENVALUE SIMULATION <==========

Bat./Gen.	k	Average k
1/1	1.43591	
2/1	1.44107	
3/1	1.46715	
4/1	1.44217	
5/1	1.44120	
6/1	1.44749	
7/1	1.41110	
8/1	1.42622	
9/1	1.41943	
10/1	1.45572	
11/1	1.42858	
12/1	1.40864	1.41861 +/- 0.00997
13/1	1.43104	1.42276 +/- 0.00710
14/1	1.39541	1.41592 +/- 0.00848
15/1	1.42233	1.41720 +/- 0.00669
16/1	1.45374	1.42329 +/- 0.00818
17/1	1.44471	1.42635 +/- 0.00756
18/1	1.43974	1.42802 +/- 0.00676
19/1	1.41935	1.42706 +/- 0.00604
20/1	1.41606	1.42596 +/- 0.00551
21/1	1.44253	1.42747 +/- 0.00521
22/1	1.39732	1.42495 +/- 0.00538
23/1	1.41900	1.42450 +/- 0.00497
24/1	1.46946	1.42771 +/- 0.00561
25/1	1.41506	1.42686 +/- 0.00529
26/1	1.37819	1.42382 +/- 0.00581
27/1	1.44714	1.42519 +/- 0.00563
28/1	1.41335	1.42454 +/- 0.00535
29/1	1.40607	1.42356 +/- 0.00515
30/1	1.41837	1.42330 +/- 0.00489
31/1	1.44305	1.42424 +/- 0.00475
32/1	1.41204	1.42369 +/- 0.00456
33/1	1.43214	1.42406 +/- 0.00437
34/1	1.41634	1.42374 +/- 0.00420
35/1	1.41686	1.42346 +/- 0.00404
36/1	1.43228	1.42380 +/- 0.00389
37/1	1.40613	1.42315 +/- 0.00380
38/1	1.44813	1.42404 +/- 0.00377
39/1	1.41006	1.42356 +/- 0.00367
40/1	1.42911	1.42374 +/- 0.00355
41/1	1.43486	1.42410 +/- 0.00345
42/1	1.41844	1.42392 +/- 0.00335
43/1	1.43858	1.42437 +/- 0.00328
44/1	1.43403	1.42465 +/- 0.00319

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45/1    1.44357    1.42519 +/- 0.00314
46/1    1.46074    1.42618 +/- 0.00321
47/1    1.45438    1.42694 +/- 0.00322
48/1    1.43480    1.42715 +/- 0.00314
49/1    1.41601    1.42686 +/- 0.00307
50/1    1.44089    1.42721 +/- 0.00301
51/1    1.44526    1.42765 +/- 0.00297
52/1    1.46619    1.42857 +/- 0.00304
53/1    1.41549    1.42827 +/- 0.00298
54/1    1.44611    1.42867 +/- 0.00294
55/1    1.41302    1.42832 +/- 0.00290
56/1    1.44250    1.42863 +/- 0.00285
57/1    1.42996    1.42866 +/- 0.00279
58/1    1.44384    1.42898 +/- 0.00275
59/1    1.43925    1.42919 +/- 0.00270
60/1    1.41463    1.42890 +/- 0.00266
61/1    1.46786    1.42966 +/- 0.00272
62/1    1.43317    1.42973 +/- 0.00267
63/1    1.39025    1.42898 +/- 0.00272
64/1    1.39920    1.42843 +/- 0.00273
65/1    1.43297    1.42851 +/- 0.00268
66/1    1.41978    1.42836 +/- 0.00263
67/1    1.42637    1.42832 +/- 0.00259
68/1    1.41588    1.42811 +/- 0.00255
69/1    1.42864    1.42812 +/- 0.00251
70/1    1.42365    1.42804 +/- 0.00247
71/1    1.38925    1.42741 +/- 0.00251
72/1    1.44995    1.42777 +/- 0.00249
73/1    1.43179    1.42783 +/- 0.00245
74/1    1.42060    1.42772 +/- 0.00242
75/1    1.41430    1.42751 +/- 0.00239
76/1    1.45173    1.42788 +/- 0.00238
77/1    1.42549    1.42785 +/- 0.00235
78/1    1.38428    1.42721 +/- 0.00240
79/1    1.42005    1.42710 +/- 0.00237
80/1    1.38496    1.42650 +/- 0.00241
81/1    1.42432    1.42647 +/- 0.00237
82/1    1.43294    1.42656 +/- 0.00234
83/1    1.41646    1.42642 +/- 0.00231
84/1    1.43316    1.42651 +/- 0.00229
85/1    1.43223    1.42659 +/- 0.00226
86/1    1.41732    1.42647 +/- 0.00223
87/1    1.40796    1.42623 +/- 0.00221
88/1    1.41033    1.42602 +/- 0.00219
89/1    1.40321    1.42573 +/- 0.00219
90/1    1.42527    1.42573 +/- 0.00216
91/1    1.43911    1.42589 +/- 0.00214
92/1    1.41497    1.42576 +/- 0.00212
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 93/1    1.39240    1.42536 +/- 0.00213
 94/1    1.43794    1.42551 +/- 0.00211
 95/1    1.41879    1.42543 +/- 0.00208
 96/1    1.42404    1.42541 +/- 0.00206
 97/1    1.45426    1.42574 +/- 0.00206
 98/1    1.41194    1.42559 +/- 0.00205
 99/1    1.42183    1.42554 +/- 0.00202
100/1    1.38124    1.42505 +/- 0.00206
101/1    1.44377    1.42526 +/- 0.00205
102/1    1.43949    1.42541 +/- 0.00203
103/1    1.45457    1.42573 +/- 0.00203
104/1    1.41552    1.42562 +/- 0.00201
105/1    1.43778    1.42575 +/- 0.00200
106/1    1.45883    1.42609 +/- 0.00201
107/1    1.46411    1.42648 +/- 0.00202
108/1    1.45698    1.42679 +/- 0.00203
109/1    1.43190    1.42684 +/- 0.00201
110/1    1.42871    1.42686 +/- 0.00199
111/1    1.45372    1.42713 +/- 0.00199
112/1    1.39394    1.42680 +/- 0.00199
113/1    1.43256    1.42686 +/- 0.00197
114/1    1.43607    1.42695 +/- 0.00196
115/1    1.39753    1.42667 +/- 0.00196
116/1    1.45915    1.42697 +/- 0.00196
117/1    1.43013    1.42700 +/- 0.00195
118/1    1.43267    1.42706 +/- 0.00193
119/1    1.41629    1.42696 +/- 0.00191
120/1    1.40336    1.42674 +/- 0.00191
121/1    1.43310    1.42680 +/- 0.00189
122/1    1.44383    1.42695 +/- 0.00188
123/1    1.41679    1.42686 +/- 0.00187
124/1    1.41070    1.42672 +/- 0.00185
125/1    1.43399    1.42678 +/- 0.00184
126/1    1.41120    1.42665 +/- 0.00183
127/1    1.42187    1.42661 +/- 0.00181
128/1    1.41109    1.42648 +/- 0.00180
129/1    1.39111    1.42618 +/- 0.00181
130/1    1.40094    1.42597 +/- 0.00181
131/1    1.43356    1.42603 +/- 0.00180
132/1    1.45773    1.42629 +/- 0.00180
133/1    1.44726    1.42646 +/- 0.00179
134/1    1.41557    1.42637 +/- 0.00178
135/1    1.41818    1.42631 +/- 0.00177
136/1    1.41313    1.42620 +/- 0.00176
137/1    1.40462    1.42603 +/- 0.00175
138/1    1.39990    1.42583 +/- 0.00175
139/1    1.43805    1.42593 +/- 0.00174
140/1    1.42692    1.42593 +/- 0.00172
```

```
     141/1    1.45419    1.42615 +/- 0.00172
     142/1    1.42725    1.42616 +/- 0.00171
     143/1    1.43064    1.42619 +/- 0.00170
     144/1    1.46675    1.42649 +/- 0.00171
     145/1    1.44579    1.42664 +/- 0.00171
     146/1    1.43131    1.42667 +/- 0.00169
     147/1    1.41065    1.42655 +/- 0.00169
     148/1    1.37228    1.42616 +/- 0.00172
     149/1    1.45894    1.42640 +/- 0.00172
     150/1    1.44600    1.42654 +/- 0.00172
Creating state point statepoint.150.h5…

=======================>     TIMING STATISTICS     <=======================

Total time for initialization     = 1.0133e+00 seconds
  Reading cross sections          = 9.9238e-01 seconds
Total time in simulation          = 6.7174e+00 seconds
  Time in transport only          = 6.5092e+00 seconds
  Time in inactive batches        = 4.0945e-01 seconds
  Time in active batches          = 6.3079e+00 seconds
  Time synchronizing fission bank = 1.1263e-01 seconds
    Sampling source sites         = 9.2041e-02 seconds
    SEND/RECV source sites        = 2.0542e-02 seconds
  Time accumulating tallies       = 2.1719e-04 seconds
  Time writing statepoints        = 5.5020e-03 seconds
Total time for finalization       = 1.1400e-06 seconds
Total time elapsed                = 7.7377e+00 seconds
Calculation Rate (inactive)       = 122116 particles/second
Calculation Rate (active)         = 110972 particles/second

============================>     RESULTS     <============================

k-effective (Collision)     = 1.42632 +/- 0.00159
k-effective (Track-length)  = 1.42654 +/- 0.00172
k-effective (Absorption)    = 1.42811 +/- 0.00110
Combined k-effective        = 1.42755 +/- 0.00090
Leakage Fraction            = 0.00000 +/- 0.00000
```